Resolution Geometry

Conditional Rigidity of Observable Geometry

Universal Properties, Selection Principles, Hypothesis Non-Omissibility, and the Anti-Ladder

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Abstract

A dependency-ordered framework invites several distinct objections. Its imported theorems might be replaced, its hypotheses might be weakened, additional structure might be inserted between its rungs, and its order might reflect presentation rather than necessity. Resolution Geometry (RG) answers these objections by distinguishing the structures determined by a specified observation system from the principles and declarations used to select additional geometry.

We separate four achievements: existence, uniqueness in a stated class, canonical selection, and empirical certification. The observable quotient and its factorization maps are canonical relative to the law map. A fixed law-space structure descends canonically, but descent and relabeling invariance do not select that structure. Characterization theorems provide further rigidity only under their full hypotheses. Effective liftable symmetry is relative to the specified latent presentation, minimal sufficient panels require an existence condition, and nondegeneracy of an indefinite pullback requires nondegeneracy on the image tangent.

The paper retains the dependency graph of the 116-rung print edition. Its reported graph has one source, L0, height 23, and width 46. The numbering is a linear extension rather than an intrinsic order of discovery. The accompanying six-check battery is reported to pass on five seeds, with three canaries detected; these computations support their stated finite checks rather than establish universal uniqueness or completeness.

The resulting rigidity claim is conditional and precise: RG organizes canonical observable constructions and geometric characterizations while identifying the inputs, selection principles, equivalences, and residual choices on which they depend.

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1 The objections and the scope of rigidity

The following objections concern different mathematical questions.

O1 *Imports.* The imported theorems (Čencov, Gleason, Malament, Oseledets, and others) might be replaced, producing different geometry.

O2 *Gaps.* Additional structures might be inserted between two rungs, so the displayed ladder need not exhaust all possible constructions.

O3 *Hypotheses.* The named hypotheses are modelling commitments. Weaker or different sufficient conditions might serve.

O4 *Order.* The position of a rung might be an editorial decision rather than a mathematical necessity.

O5 *Framework.* A categorical, information-geometric, or Bayesian formulation might organize the same observation system differently.

No single uniqueness theorem answers all five objections. Universal factorization answers one question; a metric characterization answers another; a dependency graph answers a third.

Throughout, an observation system includes a specified latent domain X, a law map

$$F : X \longrightarrow Y,$$

and the protocols, tasks, and admissible transformations relevant to the statement under consideration. The target Y may be a product of protocol-law spaces. It need not assert the existence of a coupled law across incompatible contexts.

Define

$$x \mathcal{K} x' \quad \Longleftrightarrow \quad F(x) = F(x'), \qquad Q = X/\mathcal{K},$$

and let $\pi : X \to Q$ be the quotient map.

Definition 1.1 (Four distinct achievements)**.** Let D denote the fixed inputs and hypotheses of a claim, and let $\mathcal{C}(D)$ be its admissible candidate class.

1. *Existence:* $\mathcal{C}(D)$ is nonempty.
2. *Uniqueness:* its members are equivalent under a specified equivalence.

3. *Canonical selection:* the data determine a distinguished object or equivalence class without an additional unrecorded convention.

4. *Empirical certification:* observations, together with stated modelling assumptions, justify applying the claim to the system being studied.

If $\mathrm{Aut}(D)$ acts on $\mathcal{C}(D)$, a canonical selected member must be fixed by that action. A fixed point is a necessary symmetry condition for selection, not by itself a general proof of naturality across all systems.

The central claim of this paper is therefore relative: once inputs and selection principles are fixed, some constructions are forced and others retain explicitly described freedom.

2 Law dependence and geometric selection

Definition 2.1 (Law-defined and relabeling-invariant structures)**.** A structure is *law-defined* when it is specified on the observable law image and pulled back along F. For a covariant tensor this means

$$M = F^*\mu$$

for a specified tensor μ on a suitable smooth law space.

Such a structure is *relabeling-invariant* when μ is invariant under the declared law relabelings. These properties concern dependence and symmetry. They do not assert that μ is uniquely selected.

The selection of a law-defined structure is a separate question. It may be:

1. uniquely characterized in a stated class;

2. characterized only up to a residual family;

3. selected by an explicit declaration.

A law-defined structure can therefore also be declared. These are not mutually exclusive categories.

Proposition 2.2 (Canonical descent of a fixed tensor)**.** *Suppose that on a regular stratum*

$$F = \bar{F} \circ \pi,$$

where $\pi : S \to Q$ is a surjective submersion. For a fixed covariant tensor μ on law space, set

$$M = F^*\mu, \qquad \bar{M} = \bar{F}^*\mu.$$

Then

$$M = \pi^*\bar{M},$$

and $\bar{M}$ is the unique tensor on Q with this property.

If a declared stabilizer pair satisfies

$$F \circ \varphi = \psi \circ F, \qquad \psi^*\mu = \mu,$$

then the induced action on Q preserves $\bar{M}$.

Proof. Functoriality of pullback gives

$$F^*\mu = (\bar{F} \circ \pi)^*\mu = \pi^*(\bar{F}^*\mu).$$

Pullback by a surjective submersion is injective on covariant tensors: tangent vectors at a quotient point can be lifted at any point of its fibre. This proves uniqueness.

The stabilizer identity gives

$$\varphi^*F^*\mu = F^*\psi^*\mu = F^*\mu.$$

The induced quotient action therefore preserves $\bar{M}$. □

Proposition 2.3 (Descent does not select a metric)**.** *Law dependence and outcome-relabeling invariance do not, in general, uniquely determine a metric.*

Proof. Take

$$X = (0,1), \qquad F(p) = \text{Bernoulli}(p),$$

with identity and outcome exchange as the declared relabelings. The induced exchange is $p \mapsto 1-p$.

For every smooth positive function satisfying

$$a(p) = a(1-p),$$

the tensor

$$g_a = a(p)\, dp^2$$

is a law-defined, relabeling-invariant Riemannian metric. The kernel is diagonal, so all these metrics descend.

For example,

$$g_{\mathrm{E}} = dp^2, \qquad g_{\mathrm{F}} = \frac{dp^2}{p(1-p)}$$

are not constant multiples. Since F is injective, a reparametrization r satisfying $F \circ r = F$ must be the identity. Thus these distinct metrics are not identified by a law-preserving reparametrization. □

Corollary 2.4 (The scope of descent rigidity)**.** *Canonical descent determines the quotient structure* given the law-space structure*. It does not determine the law-space structure itself. A uniqueness claim for that structure requires a separate characterization theorem or selection principle.*

For Fisher geometry, the relevant additional requirements concern statistical morphisms across an appropriate class of experiments, not merely relabelings of one fixed outcome space. The full hypotheses of the characterization must be retained [3].

3 Canonical constructions and conditional characterizations

3.1 Universal factorization

Theorem 3.1 (Observable quotient)**.** *In the category of sets, the quotient $Q = X/\mathcal{K}$ has the following universal property. A map $H : X \to Z$ is constant on the fibres of F if and only if there is a unique map $\bar{H} : Q \to Z$ satisfying*

$$H = \bar{H} \circ \pi.$$

Moreover,

$$Q \cong \operatorname{im} F$$

canonically as sets.

Proof. Define $\bar{H}([x]) = H(x)$. Fibre constancy makes this well defined, and surjectivity of π proves uniqueness. The map $[x] \mapsto F(x)$ is a well-defined bijection onto $\operatorname{im} F$. □

Equivalently, π is the coequalizer of the kernel pair. It is the coarsest quotient of X through which F factors. The universal property is understood in its specified category [8].

A set-level quotient does not automatically carry a manifold structure. Smooth statements require the regularity assumptions used in the differential layer.

Proposition 3.2 (Joint kernels and specialization)**.** *For a product-valued law map $F = (F_i)_{i \in I}$,*

$$\mathcal{K}_F = \bigcap_{i \in I} \mathcal{K}_{F_i}.$$

If two equivalence relations on the same set satisfy

$$\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0,$$

there is a unique map

$$s : Q_\varepsilon \longrightarrow Q_0$$

such that

$$\pi_0 = s \circ \pi_\varepsilon.$$

Proof. Equality in a product is equality in every coordinate. For specialization, set $s([x]_\varepsilon) = [x]_0$. Kernel inclusion makes this well defined, and surjectivity of π_ε gives uniqueness. □

Task descent is the same factorization principle applied to a task constant on observational fibres.

3.2 Effective liftable symmetry

For the specified latent presentation and declared law transformations, let

$$\mathrm{Stab}(F) = \{(\varphi, \psi) : F \circ \varphi = \psi \circ F\}.$$

Its action on Q has an image

$$\Gamma = \mathrm{im}\big(\mathrm{Stab}(F) \to \mathrm{Aut}(Q)\big).$$

Proposition 3.3 (Presentation-relative determination)**.** *The effective liftable symmetry group* Γ *is determined by the specified observation system and its admissible transformations. Equality of observable law images alone does not determine* Γ*.*

Proof. The first assertion follows from the definition. For the second, consider

$$F_A(x) = \mathcal{N}(x, 1), \qquad X_A = \mathbb{R}.$$

Every translation of the mean lifts to a translation of X_A.

Now take

$$X_B = (\mathbb{R} \times \{0\}) \sqcup ((0, \infty) \times \{1\}), \qquad F_B(x, j) = \mathcal{N}(x, 1).$$

Both law images are the full Gaussian location family. The fibres of F_B have cardinality one for means $q \leq 0$ and two for means $q > 0$.

A lift of the translation $q \mapsto q + a$ would induce bijections between corresponding fibres. It must therefore preserve the positive half-line and its complement, which forces $a = 0$. Thus the liftable translation subgroup is all of $\mathbb{R}$ for A and trivial for B. □

Presentation invariance is a further claim. It requires an equivalence of observation systems that transports the relevant actions. It does not follow from equality of law images.

3.3 Finite minimal-panel antichains

Theorem 3.4 (Finite-panel characterization)**.** *Let* I *be finite, and let* $\mathcal{S} \subseteq 2^I$ *be an upward-closed family of sufficient panels. Its inclusion-minimal members form a uniquely determined antichain* $\mathcal{M}$*, and*

$$\mathcal{S} = \{S \subseteq I : \exists M \in \mathcal{M},\ M \subseteq S\}.$$

Proof. Every sufficient panel contains a minimal sufficient subpanel, because its collection of subpanels is finite. Distinct minimal members cannot contain one another. The displayed identity follows from minimality and upward closure, and it determines $\mathcal{M}$ uniquely. □

For infinite I, existence of minimal members needs an additional assumption. For example, let

$$X = \{0, 1\}^{\mathbb{N}}, \qquad F_n(x) = (x_1, \ldots, x_n), \qquad T(x) = x.$$

A panel is sufficient exactly when its indices are unbounded. Removing any one index preserves sufficiency. There are sufficient panels but no inclusion-minimal ones.

3.4 Efficient information and minimal-norm lifts

These are conditional characterizations, not consequences of quotient factorization alone.

For a regular finite-dimensional model with information matrix

$$I = \begin{pmatrix} I_{\theta\theta} & I_{\theta\eta} \\ I_{\eta\theta} & I_{\eta\eta} \end{pmatrix},$$

and invertible nuisance block $I_{\eta\eta}$, efficient information is

$$I_{\text{eff}} = I_{\theta\theta} - I_{\theta\eta} I_{\eta\eta}^{-1} I_{\eta\theta}.$$

Its interpretation requires the stated regular statistical model and nuisance convention. Positive definiteness is not automatic.

Likewise, if a surjective linear map $A : E \to B$ is equipped with a positive-definite inner product on E, each affine fibre $A^{-1}(b)$ has a unique minimal-norm member. This selects a horizontal lift relative to the declared inner product. It does not select that inner product.

Why a groupoid? Equality of laws determines an equivalence-relation groupoid, with one arrow between each pair of equivalent points. This is a canonical encoding of the equivalence relation. A quotient, a kernel pair, and an equivalence-relation groupoid can express the same factorization content. The choice of language is not itself a uniqueness theorem for additional geometry.

4 The differential layer and nondegeneracy

Theorem 4.1 (Radical of a pulled-back symmetric tensor)**.** *Let $\pi : S \to Q$ be a surjective submersion, let $\bar{F} : Q \to Y$ be an immersion, and set*

$$F = \bar{F} \circ \pi, \qquad V_x = \ker D\pi_x.$$

For a symmetric covariant two-tensor μ on Y, define

$$\bar{M} = \bar{F}^* \mu, \qquad M = F^* \mu.$$

Then

$$\operatorname{rad} M_x = (D\pi_x)^{-1}\big(\operatorname{rad} \bar{M}_{\pi(x)}\big).$$

Consequently,

$$\operatorname{rad} M_x = V_x$$

if and only if $\bar{M}_{\pi(x)}$ is nondegenerate.

Proof. For $u, v \in T_x S$,

$$M_x(u, v) = \bar{M}_{\pi(x)}(D\pi_x u, D\pi_x v).$$

Since $D\pi_x$ is surjective, this vanishes for every v exactly when $D\pi_x u$ lies in the radical of $\bar{M}$. The final assertion follows immediately. □

Writing

$$W_q = D\bar{F}_q(T_q Q),$$

the required condition is nondegeneracy of $\mu|_{W_q}$. If μ is an ambient nondegenerate bilinear form, this is equivalent to

$$W_q \cap W_q^{\perp_\mu} = \{0\}.$$

Positive definiteness of μ guarantees this condition. Ambient nondegeneracy alone does not.

For example, on $\mathbb{R}^2$ let

$$\mu = du^2 - dv^2, \qquad \bar{F}(t) = (t, t).$$

Then μ is nondegenerate and $\bar{F}$ is immersive, but

$$\bar{F}^* \mu = 0.$$

Thus the positive-definite branch and the indefinite branch require different nondegeneracy statements.

5 Imports by logical type

Imported theorems have different roles. Some characterize a structure; others establish existence, regularity, inequalities, representations, or obstructions. Their conclusions are conditional on their full hypotheses.

The following table retains the rung references of the 116-rung print edition. It summarizes logical roles rather than replacing the complete statements in that edition.

rung	logical role	import or construction	required scope	residual or limitation
L28	characterization	Čencov: Fisher up to a positive scale	compatible metric family on finite probability simplices; the full statistical-morphism hypotheses	scale; normalization must be stated
L21, L26	regularity	slice theorems, semicontinuity, Hardt-type results	the applicable properness, regularity, or tameness assumptions	no uniqueness of all stratifications is implied
L29	reconstruction	regular second-order reconstruction	the reconstruction hypotheses stated in the print edition	only the data and equivalence covered by that result
L34	inequality	data processing for the selected distinguishability structure	the relevant metric or divergence and admissible Markov maps	not implied by kernel inclusion alone
L37	family characterization	Petz monotone metrics	the specified quantum statistical setting and CPTP monotonicity	operator-monotone function f; potentially function-valued freedom
L41	probability representation	Gleason and Busch trace-rule results	the relevant event space, dimension, and additivity assumptions	a state is represented, not selected without data
L42	compatibility and obstruction	Fine-type joint-law characterization	the specified Bell scenario and consistency conditions	not a universal joint-law theorem for arbitrary contexts
L45	finite-sample reconstruction	the reconstruction result used in the print edition	model, sample size, error criterion, and boundary assumptions	a certificate with stated coverage or error
L47, L49	asymptotic representation	Bernstein–von Mises and LAN	identifiability, regularity, prior, and asymptotic assumptions	no unconditional washing out of priors
L54	bounds and certification	Le Cam two-point bounds; DKW	the relevant sampling model and discrepancy	DKW controls CDF error, not unrestricted TV error
L56	measure-theoretic structure	pairwise domination; Feldman–Hájek	pairwise comparison; Gaussian assumptions for the dichotomy	no Fisher layer on arbitrary singular directions
L65	geometric characterization	conformal class from a regular quadratic null cone	appropriate dimension and nondegenerate Lorentzian setting	conformal factor remains
L67, L69, L70	regularity, duality, and obstruction	Hardt-type results, cone duality, Gårding theory	tameness, specified duality, and hyperbolicity assumptions	not every characteristic set is a quadratic cone

rung	logical role	import or construction	required scope	residual or limitation
L74	connection construction	Levi-Civita connection or specified projected construction	a fixed nondegenerate metric and the stated regularity	projection needs its own nondegeneracy assumptions
L82	local representation	Baker–Campbell–Hausdorff	Lie structure and a formal or convergent local setting	not an unrestricted global composition formula
L84, L85, L87	entropy results and criteria	Misiurewicz–Szlenk, Ruelle, Pesin, cone-field criteria	their respective dynamical hypotheses	inequalities and equalities must remain distinct
L88, L90	asymptotic representation	Oseledets and Brudno	integrability, invariant measure, and relevant ergodicity assumptions	protocol and coding commitments remain explicit
L95	scale characterization	additive representation and ratio-scale results	the stated concatenation, order, and regularity assumptions	unit or calibration
L99	rigidity	Beckman–Quarles	the theorem's Euclidean domain and $n \geq 2$	not an arbitrary-domain isometry theorem
L102	stationary representation	Landau–Lifshitz spatial or radar construction	the specified stationary setting and operational convention	non-stationary extension requires separate treatment

Proposition 5.1 (Conditional import rigidity)**.** *Suppose an imported theorem characterizes a unique equivalence class in a candidate class $\mathcal{C}$ under assumptions A. Any alternative in $\mathcal{C}$ satisfying the same assumptions belongs to that equivalence class.*

If the theorem instead classifies a family, an additional selection rule is needed to choose a member unless the application retains the family.

Proof. The first assertion is the characterization conclusion. The second follows because classification of a family does not distinguish one of its members. □

This proposition does not say that every imported theorem is a characterization theorem. Nor does it show that the assumptions A are the only possible sufficient assumptions.

Remark 5.2 (Kernel inclusion is not metric monotonicity). For a Markov garbling M,

$$\mathcal{K}_F \subseteq \mathcal{K}_{MF}.$$

This concerns exact equality of laws. It does not imply contraction of an arbitrary metric.

For example, take

$$p = (0.3, 0.3, 0.4), \qquad q = (0.2, 0.2, 0.6),$$

and merge the first two outcomes. The squared Euclidean distance changes from

$$\|p - q\|_2^2 = 0.06$$

to

$$\|Mp - Mq\|_2^2 = 0.08.$$

Metric monotonicity is therefore a separate requirement or a separately proved property of the chosen construction.

6 Hypothesis non-omissibility

Definition 6.1 (Non-omissibility)**.** For a theorem

$$B \wedge H \Longrightarrow C,$$

a countermodel satisfying

$$B \wedge \neg H \wedge \neg C$$

shows that H cannot simply be deleted while retaining B. We call this *non-omissibility relative to* B.

Such a countermodel does not establish

$$B \wedge C \Longrightarrow H,$$

and does not exclude a weaker sufficient condition H'.

The following table retains the sixteen countermodel targets reported for UM04. A necessity claim stronger than non-omissibility would require a separate converse theorem.

hypothesis	reported countermodel	conclusion tested
H-unif	moving bump: pointwise convergence while an identified pair separates	kernel specialization under convergence
H-hell	$\mathcal{N}(\varepsilon y, \varepsilon^2)$: Hellinger separation persists while means converge	distinguishability specialization
H-adm	concentrating densities retain TV $\geq \frac{3}{4}$ after transfer	uniform forgetting over a class
H-cov	a vertical shear preserves the law but changes a lift	selection of a latent connection
H-SI	dilation of (x, c, L) preserves two delays	absolute length calibration
H-exch	a doubling-block record has oscillating frequencies	the stated frequency-to-probability inference
H-task	SLD, BKM, and WY differ on one tangent	selection of a quantum metric
H-med	opposite temperature gradients yield equal delays	medium reconstruction
H-reg	a fold has vanishing first-order information at the wall	the regular differential description
H-nuis	efficient information tends to zero near nuisance collinearity	nondegenerate efficient geometry
H-coh	completeness holds only on a selected subset	extension of completeness across a stratum
H-sound	an unsound arrow merges distinct laws	sound quotient comparison
H-int	a scale cocycle has product 1.25	existence of a global scale representative
H-stat	drift changes a transferred scale	the stated stationary or ergodic transfer
H-Γ	rotations without boosts leave two invariant forms on $1+3$	Lorentz-form uniqueness up to scale
H-val	a nonlinear detector passes one anchor but fails elsewhere	extension of anchor calibration

For each target, the relevant proof obligation is to verify the remaining premises B, failure of H, and failure of C. A numerical witness supports only the properties actually checked.

Remark 6.2 (Theorem assumptions and certification assumptions). A convergence theorem may require locally uniform convergence, while a finite-data certificate additionally requires a known modulus or another uniform control. These are different assumptions.

Similarly, a failure of nondegeneracy at a limit does not by itself establish discontinuity. The conclusion tested by a countermodel must be stated exactly.

7 Empirical certification of exact structure

The quotient is exact relative to a specified law map. Its empirical certification is a separate statistical problem.

Proposition 7.1 (No unrestricted finite-sample equality certificate)**.** *Fix a sample size n. Over Bernoulli alternatives arbitrarily close to* Bernoulli$(1/2)$*, no test uniformly distinguishes exact equality from every nonzero perturbation with both error probabilities bounded by a fixed $\alpha < 1/2$.*

Proof. Let

$$P_0 = \text{Bernoulli}(1/2), \qquad P_\delta = \text{Bernoulli}(1/2 + \delta),$$

where $0 < |\delta| < 1/2$. A coordinatewise coupling gives

$$\text{TV}(P_0^{\otimes n}, P_\delta^{\otimes n}) \le n|\delta|.$$

For any binary test ϕ,

$$P_0^{\otimes n}(\phi = 1) + P_\delta^{\otimes n}(\phi = 0) \ge 1 - n|\delta|.$$

If both errors were at most α, choosing $n|\delta| < 1 - 2\alpha$ would give a contradiction. □

This does not prevent exact conclusions from a certified structural model, a separated hypothesis class, or other additional information. It prevents identifying finite-sample indistinguishability with exact law equality without such assumptions.

Accordingly, applications should distinguish:

1. the model-level law map;
2. its exact kernel and quotient;
3. the confidence correspondence or error certificate available from observations;
4. the task conclusion justified by that certificate.

8 The anti-ladder: the dependency graph of the displayed proofs

The graph statistics reported for this edition are:

- 116 rungs and 297 dependency edges;
- 192 edges in the transitive reduction;
- one source, L0;
- height 23 and width 46.

Here height counts vertices in a longest chain, and width is the maximum antichain size. These are edition-specific computational results.

Proposition 8.1 (Order represented by an acyclic dependency graph)**.** *For a finite directed acyclic graph, reachability defines a partial order after adjoining equality. A numbering is a linear extension exactly when every edge points from a lower-numbered to a higher-numbered vertex.*

If the graph has a unique source, every vertex is reachable from that source.

Proof. Acyclicity gives antisymmetry of reachability; reflexivity and transitivity follow by construction. The numbering criterion is the definition of a topological ordering.

Tracing predecessors from any vertex in a finite acyclic graph eventually reaches a source. If there is only one source, that source reaches every vertex. □

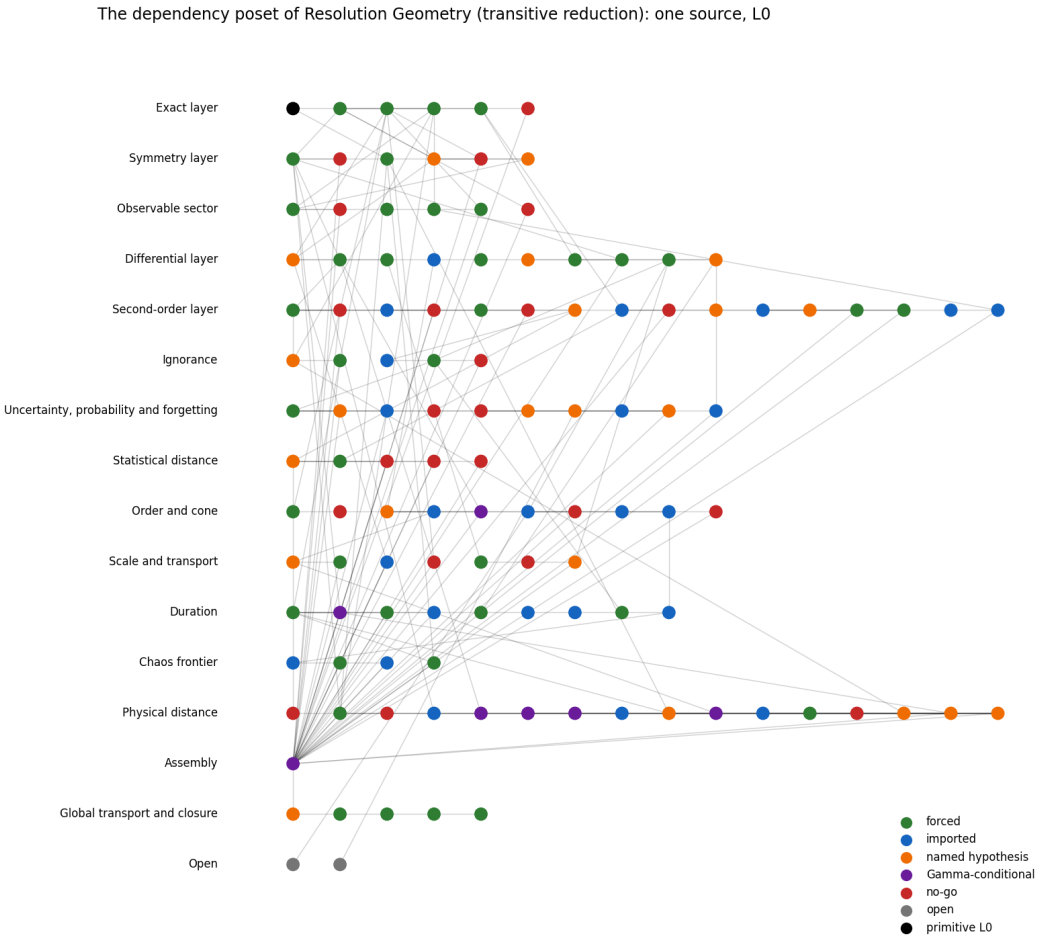


Figure 1: The supplied dependency graph of the 116-rung print edition, shown through its reported transitive reduction with 192 essential edges and coloured by the edition's status labels. The graph records the displayed organization of definitions and proofs. A "forced" label is read relative to the inputs and hypotheses of the corresponding rung.

The reported longest chain is represented in the edition by

$$\begin{aligned} &\mathrm{L0} \to \mathrm{L1} \to \mathrm{L2} \to \mathrm{L18} \to \mathrm{L21} \to \cdots \to \mathrm{L27} \\ &\to \mathrm{L83} \to \cdots \to \mathrm{L90} \to \mathrm{L108} \to \cdots \to \mathrm{L113}. \end{aligned}$$

The ellipses abbreviate the chain recorded by the graph data.

Proposition 8.2 (Positions are presentation)**.** *Let P be a finite poset with n elements. Write $a(v)$ and $d(v)$ for the numbers of strict ancestors and strict descendants of v. In zero-based indexing, the earliest and latest positions of v among linear extensions are*

$$a(v) \qquad \text{and} \qquad n-1-d(v).$$

Proof. All ancestors must precede v, and all descendants must follow it.

The ancestor set is an order ideal. Place a linear extension of that ideal first, then v, then extend the remaining order. This attains position $a(v)$.

Dually, the descendants form an order filter. Place the remaining elements first, then v, then its descendants. This attains position $n-1-d(v)$. □

The reported median positional slack is 93. Only L0 has a fixed position in this edition; the reported imported-rung slack ranges from 59 to 105. Counting linear extensions of a general finite poset is #P-complete [7]; the reported slack calculation does not require enumerating them.

Remark 8.3 (Proof dependencies are not a minimal axiomatization)*.* The graph records the definitions and proof routes displayed in the edition. Another proof can remove or replace proof dependencies. A different axiomatization can organize the same results with a different graph.

Consequently, the graph establishes neither globally minimal assumptions nor a uniquely necessary proof order. Renumbering preserves the represented order; changing proofs can change it.

Remark 8.4 (Cycle canaries)*.* Adding an edge $v \to u$ creates a cycle precisely when a path $u \rightsquigarrow v$ already exists. A numerically backward edge need not create a cycle. It may instead invalidate the displayed numbering as a linear extension. A graph canary must state which property it tests.

9 The rigidity contract

A claim that a structure is forced should specify:

1. the fixed inputs;
2. the candidate class;
3. the selection or characterization requirement;
4. the permitted equivalence;
5. the residual freedom;
6. the certification available in applications.

This is a reporting discipline, not an additional uniqueness theorem.

structure	fixed inputs and class	rigidity statement	residual and certification
Observable quotient	specified X and F; set quotients	universal factorization	no set-level residual; empirical kernel certification is separate
Effective symmetry	latent presentation and admissible transformations	image of the stabilizer action	presentation-relative; transport requires a system equivalence

structure	fixed inputs and class	rigidity statement	residual and certification
Descended tensor	fixed μ and regular quotient	unique tensor with the stated pullback	choice of μ remains unless characterized
Fisher metric	compatible metric family in the characterization's domain	uniqueness up to positive scale	normalization and applicability of the statistical model
Quantum metric	the specified Petz class and task criterion	family characterization, followed by any justified selection	function-valued freedom unless the criterion identifies a member
Minimal panels	finite protocol set and fixed task	unique minimal generating antichain	ties retained; exact sufficiency needs a model or certificate
Specialization	kernel inclusion on a common domain	unique quotient factorization map	certification of the inclusion is separate
Latent lift	surjective differential and positive-definite latent metric	unique minimal-norm lift	the latent metric is declared or independently characterized
Cone geometry	specified cone and candidate quadratic forms	conformal characterization under the stated assumptions	scale and nonquadratic alternatives outside the class
Physical scale	counted channel and additive representation assumptions	ratio-scale characterization	unit and operational calibration
Uniform forgetting	specified admissible class and operator	a uniform bound over that class	class choice; finite examples do not certify the supremum

10 What is and is not forced

The following distinctions answer the original objections.

Canonical relative to fixed inputs. The set quotient, task factorization, joint-kernel intersection, and specialization under kernel inclusion have exact factorization statements. A fixed tensor descends under the stated regularity conditions.

Characterized under additional requirements. Fisher geometry, quantum monotone metric families, scale representations, and other imported structures have the rigidity supplied by their particular characterization theorems. Their candidate classes and hypotheses are part of the claim.

Declared or residually selected. Protocols, tasks, admissible classes, latent covariance, nuisance conventions, anchors, subgroups, and target precision may be inputs. Their declaration makes the dependence explicit; it does not make the inputs consequences of the law map.

Presentation. Rung numbering, section grouping, notation, and proof routes are not uniquely forced. The supplied graph describes one edition's proof organization.

Normalization. A positive constant rescaling may preserve kernels, orthogonality, or other scale-insensitive statements. It changes numerical lengths and thresholds unless the corresponding conventions are transformed. Normalization is therefore harmless only for claims invariant under that rescaling.

No general claim is made that these lists exhaust every possible construction between rungs. Exhaustiveness of a particular audit must be demonstrated at the level of definitions and proofs, not inferred from the presence of status labels.

11 Anticipated objections

"Why Fisher rather than another law-defined metric?" Relabeling invariance alone permits many metrics (Proposition 2.3). The Fisher characterization uses stronger compatibility requirements across statistical experiments. A Wasserstein construction additionally uses outcome geometry. It can be appropriate when that geometry is part of the specified protocol, but it is not thereby the same characterization problem.

"Different hypotheses could give different geometry." Yes. A countermodel to deleting a hypothesis establishes non-omissibility relative to the remaining premises, not uniqueness of the hypothesis package. An alternative sufficient package is a legitimate mathematical proposal and should be compared by its conclusions, domain, and empirical justification.

"The imports contain the content." The imported results retain their own mathematical content. RG's assembly contributes the explicit identification of observable equivalence, compatible constructions, dependencies, and failure conditions. Replacing an import is permissible if the replacement proves the required conclusion under assumptions that are stated and justified. A characterization theorem constrains replacements only within its actual scope.

"A categorical formulation would differ." A formulation representing the same universal problem in an equivalent category recovers the corresponding universal object up to the appropriate isomorphism. Merely representing the same law image does not establish equivalence of latent presentations or liftable symmetry groups.

"A Bayesian formulation would select the missing structure." A Bayesian formulation adds a prior and a likelihood model. If the likelihood is constant along an observational fibre, the data do not distinguish points within that fibre. Under suitable regularity and identifiability assumptions, asymptotic results can describe concentration on observable parameters. They do not imply unconditional prior independence, nor do they select a unique prior along unresolved fibres.

"Do the finite checks prove rigidity?" No. They verify finite instances, numerical identities, parser conditions, and specified failure witnesses. Universal conclusions require mathematical arguments. Canaries show sensitivity to their injected faults, not completeness against every possible fault.

12 Conclusion

Resolution Geometry does not require its inputs to select themselves. Its defensible rigidity claim is more precise: the observable quotient is canonical relative to the law map; fixed law-space structures descend canonically; and additional geometry is characterized only under explicitly stated selection principles and regularity assumptions.

This separates genuine freedom from genuine necessity. Law dependence is not selection. Liftable symmetry is relative to the specified presentation. Finite-panel minimality is not an unrestricted existence theorem. Indefinite pullbacks require nondegeneracy on the image tangent. Hypothesis countermodels establish non-omissibility rather than a universally necessary hypothesis package. Exact model structure and finite-data certification remain distinct.

The dependency graph retains its value as a map of the displayed proofs. Its source, chains, antichains, and positional slack describe that organization without claiming a globally minimal axiomatization.

Read in this form, rigidity is neither an assertion that all alternatives are impossible nor a retreat to arbitrary modelling. It is a collection of explicit contracts: fixed inputs, candidate classes,

requirements, equivalences, residual freedoms, and certificates. Those contracts identify exactly what an alternative must preserve, what it may change, and where a genuine counterexample would apply.

A Reproducibility and interpretation of the reported checks

The supplied reproducibility record reports that

```
rg_rigidity_attack.py -tex "Resolution Geometry.tex"
```

passes UM01–UM06 on seeds

$$20261021, \quad 1, \quad 42, \quad 777, \quad 2026.$$

It also reports that `-figs` writes the dependency figure and that three canaries are detected.

These are reported results for the supplied edition. They are not a claim that the revised manuscript has been re-executed against an updated battery.

A.1 Scope of the six checks

UM01: finite quotient checks. The record uses all 52 partitions of a five-point set and 120 random law assignments. For each assignment, it checks the coarsest factoring partition against the kernel partition. The universal statement is Theorem 3.1; the computation checks finite instances.

UM02–UM03: geometric examples. The record reports Fisher and Hellinger infinitesimal invariance under tested Markov embeddings, contrasting Euclidean and inverse-square constructions. The factor $\frac{1}{4}$ refers to the specified infinitesimal Hellinger convention, not to equality of every global Hellinger distance with a Fisher distance.

It also reports invariant-form dimensions for specified representations, trace-rule normalization examples, minimal-norm lift comparisons, and distinct SLD, BKM, and WY values on a tested tangent. Representation, normalization, and task conventions are part of these checks.

UM04: hypothesis countermodels. The sixteen targets are listed in Section 6. Each witness must verify the remaining premises as well as the failed conclusion. Numerical examples do not by themselves prove that no weaker sufficient hypothesis exists.

UM05: dependency graph. The record supplies the graph statistics and positional slack used in Section 8. These describe the parsed edition and its recorded edges.

UM06: status and hypothesis parsing. The record reports a typed status for each of the 116 rungs and definitions for hypothesis names appearing in those statuses. A parser verifies syntactic coverage. It does not prove that every semantic assumption in every proof has been recorded.

A.2 Canaries

The supplied record lists:

1. substitution of the discrete partition for the coarsest factoring kernel partition;
2. forced coincidence of distinct Petz-family examples;
3. injection of a backward dependency into the rung table.

The first canary must use a law assignment with a nontrivial kernel if the discrete partition is to be an incorrect answer. The third must distinguish a cycle test from a numbering-consistency test, as explained in Section 8.

A.3 Recorded implementation refinements

The supplied record describes the following refinements:

1. literal arithmetic in several countermodel checks was replaced by computations;
2. an edge-reversal order test was replaced by positional-slack analysis;
3. the backward-dependency injection was changed so that successful modification of the intended row is asserted.

These refinements concern test implementation. Their mathematical interpretation remains limited to the properties actually checked.

A.4 Additional analytic regression targets

The present statements identify further tests that an updated battery should include:

1. distinct relabeling-invariant Bernoulli metrics with the same exact kernel;
2. equal Gaussian law images with different liftable translation subgroups;
3. an indefinite ambient form with a null one-dimensional image tangent;
4. an infinite sufficient-panel family without minimal members;
5. a garbling that increases Euclidean distance while preserving kernel inclusion;
6. finite-sample alternatives with distinct exact kernels and arbitrarily small statistical separation.

The proofs and constructions in the main text establish these examples. No execution result for an additional battery is asserted here.

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